CS 103000
Prof. Madeline Blount

Week 8:
REVIEW & MIDTERM!

Dall-E 2: cats learning C++ in the forest on '90's technology

MIDTERM!

💬 PRACTICE: TEXT ANALYSIS

Write a program that counts the number of each vowel in this text:

“So died these men as became Athenians. You, their survivors, must determine to have as unfaltering a resolution in the field, though you may pray that it may have a happier issue. And not contented with ideas derived only from words of the advantages which are bound up with the defense of your country, though these would furnish a valuable text to a speaker even before an audience so alive to them as the present, you must yourselves realize the power of Athens, and feed your eyes upon her from day to day, till love of her fills your hearts; and then, when all her greatness shall break upon you, you must reflect that it was by courage, sense of duty, and a keen feeling of honor in action that men were enabled to win all this, and that no personal failure in an enterprise could make them consent to deprive their country of their valor, but they laid it at her feet as the most glorious contribution that they could offer.”

Also tell me how many alphabetic characters this string has. Finally, print the text out at the end, with every vowel replaced with an “@” sign.

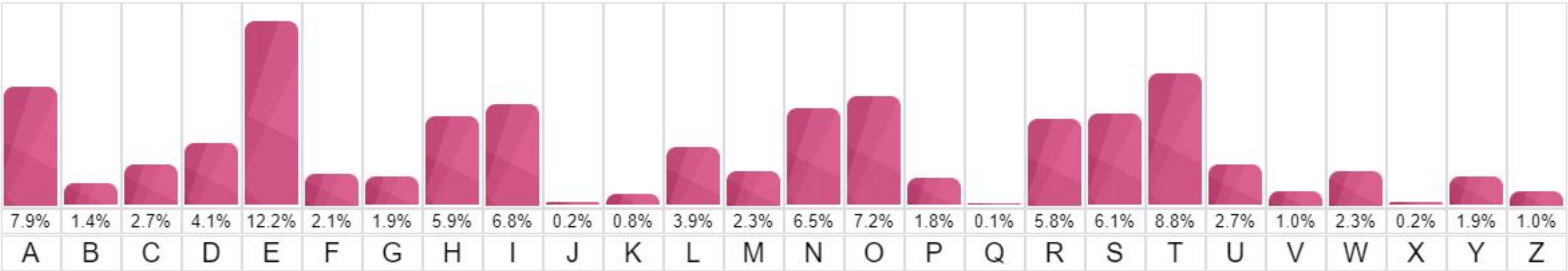
💬 **PRACTICE:**

- a: 61
- e: 102
- i: 40
- o: 60
- u: 36

743 alphabetic characters.

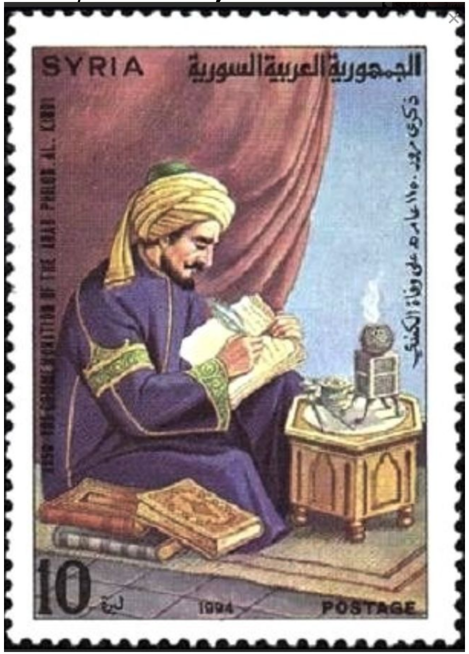
📡 CRYPTOGRAPHY: FREQUENCY ANALYSIS AND THE ONE-TIME PAD

How to hack a Caesar cipher?

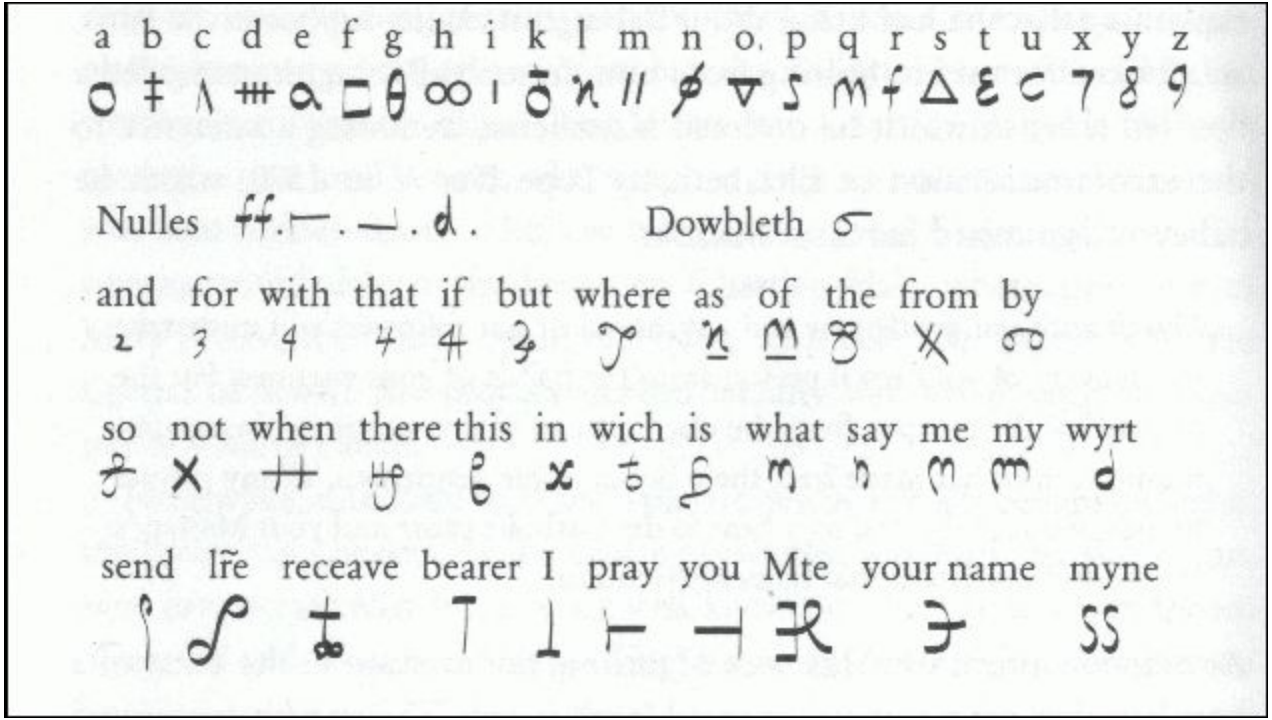


📡 CRYPTOGRAPHY: FREQUENCY ANALYSIS AND THE ONE-TIME PAD

"On Decrypting Encrypted Correspondence" - Al-Kindi, Baghdad, 850 CE



- Look @ encrypted text, count frequencies
- Which letter comes up the most? Compare that to "fingerprint" of frequencies you already know about your language …
- Famously, Mary Queen of Scots beheaded, 1587 because linguist under Queen Elizabeth (Thomas Phelippes) cracked code using frequency analysis



Mary Queen of Scots - plot w/Anthony Babington

https://www.simonsingh.net/The_Black_Chamber/maryqueenofscots.html

📡 CRYPTOGRAPHY: FREQUENCY ANALYSIS AND THE ONE-TIME PAD

- With Caesar shift, only 26 possibilities to crack
- If *every* letter was shifted by its own shift, based on a random number … 26 * 26 * 26 …
- **Gets large very quickly**
- One-time pad (1880s), pad of paper
  (or similar - 🔎 small, flammable, etc.)

🤖 Mid-term hackathon

- Work today, in-class Wednesday, in-class Friday, and any time that you choose in between

- Delegate + discuss with team!

- Research and ask questions!

- Research *concepts* not code!

🤖 <u>**Mid-term hackathon**</u>

- You may not
  - Use methods from outside of class <u>***without citation***</u>
  - Examples:
    - using #include <algorithm>
    - using .reverse, .begin(), .end()
    - using .map()
    - using &, *
  - Programs should include in comments ***why*** you chose this method, vs. another method we've studied
  - Solutions that use such methods <u>**without citation/explanation**</u> will lose points

🚀 Grades

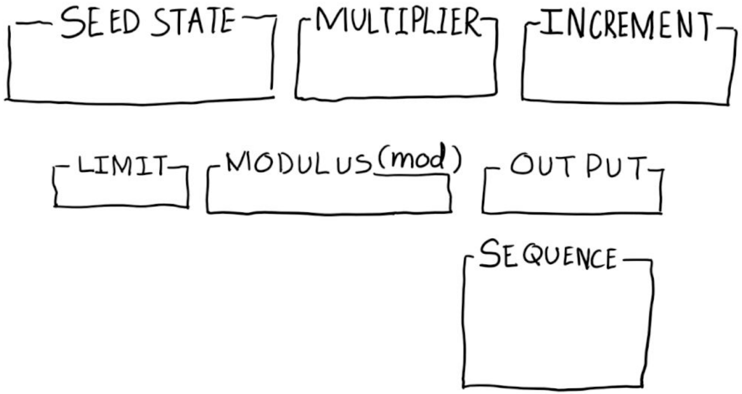
- zyBooks 16.40 will accept any submission! No “test” feedback this time
- I will be grading your code by hand 👀
- Individual grade, self-assessment
- Mid-term = 20% (same as all reading!)
- If helps class average, I will curve
- After mid-term, will learn current class grade in Brightspace
- Extra credit opportunities! 🤖

pseudorandom!

- rand() = Linear Congruential Generator (LCG)

- x = ((a * x) + c) % m

- Next number based on the previous (**state**)



`srand() = SEED`